

Groups, rings and modules

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Lecture 1 – 22/01/26

§1 Groups – a reminder

Definition 1.1. A *group* is a triple $(G, \cdot, e)$ where G is a set,

$$\cdot : G \times G \rightarrow G$$

is a function, and $e \in G$ such that

- *Associativity.* $(a \cdot b) \cdot c = a \cdot (b \cdot c)$;
- *Identity.* For all $a \in G$, $a \cdot e = e \cdot a = a$;
- *Inverses.* For all $a \in G$, there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

The size of G , $|G|$, is called the *order* of the group.

Proposition 1.2 (Uniqueness of inverses)

Inverses are unique.

Proof. Left as an exercise (see 1A). ■

Definition 1.3 (Subgroups). If G is a group and $H \subseteq G$, then we say H is a *subgroup* if

- $e \in H$;
- for all $a, b \in H$, $a \cdot b \in H$;
- $(H, \cdot, e)$ is a group.

In order to say item 3, we must first establish 1 and 2.

Proposition 1.4

A non-empty subset $H \subseteq G$ is a subgroup if and only if, for all $h_1, h_2 \in H$, $h_1 h_2^{-1} \in H$.

Proof. Left as an exercise (see 1A). ■

Definition 1.5. A group G is called *abelian* if, for all $g_1, g_2 \in G$, $g_1 g_2 = g_2 g_1$.

Example 1.6 • We have $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}/n\mathbb{Z}$.

- The symmetric groups S_n : permutations of $\{1, \dots, n\}$ under composition. More generally for any set X , we have $\text{Sym}(X)$.
- The dihedral groups D_{2n} : the group of symmetries of a regular n -gon.
- The general linear group over $\mathbb{R}$, $\text{GL}(n, \mathbb{R})$ or $\text{GL}_n(\mathbb{R})$, is the set of $\{f : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid f \text{ is an } \mathbb{R}\text{-linear bijection}\}$. Since $\text{GL}(n, \mathbb{R})$ is made of functions, we can think of it a subset of $\text{Sym}(\mathbb{R}^n)$. (Equivalently, it's the set of n by n invertible matrices with elements in $\mathbb{R}$).

Some examples of subgroups:

- The even permutations $A_n \leq S_n$ (the alternating group);
- Rotational symmetries of an n -gon, $C_n \leq D_{2n}$;
- Determinant one matrices $\text{SL}(n, \mathbb{R}) \leq \text{GL}(n, \mathbb{R})$.

Other examples include $C_2 \times C_2$, $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.

Definition 1.7. Let G be a group, $g \in G$ and $H \leq G$ a subgroup. Then the *left coset*

$$gH = \{gh : h \in H\}$$

Basic properties include that $g \in gH$, so cosets are non-empty. Also, cosets partition G – they have no non-trivial overlaps and every g lies in some coset. Furthermore, the map

$$\begin{aligned} H &\rightarrow gH \\ h &\mapsto gh \end{aligned}$$

defines a bijection.

Theorem 1.8 (Lagrange)

If G is a finite group and $H \leq G$ then

$$|G| = |H| \cdot |G : H|$$

where $|G : H|$, the index of H in G , is the number of cosets of H in G .

Definition 1.9. Let G be a group and let $g \in G$. The *order* of g , $\text{ord}(g)$, is the smallest positive integer n such that $g^n = e$, or infinity if no such n exists.

Proposition 1.10

Let $g \in G$. Then, $\text{ord}(g) \mid |G|$.

Proof. Let $g \in G$ and consider $H = \{e, g, g^2, \dots, g^{n-1}\}$ where $n = \text{ord}(g)$. This is a subgroup: it is non-empty, and $g^r \cdot g^{-s} = g^{r-s} \in H$ (after potentially multiplying by $g^n = e$). Also, the elements of H are distinct – if $g^i = g^j$ for $0 \leq i < j \leq n$, then $g^{j-i} = e$, contradicting the minimality of $\text{ord}(g)$. Hence $|H| = \text{ord}(g)$ – now apply Lagrange. ■

§2 Normal subgroups, quotients and homomorphisms

Let $H \leq G$. When are two cosets of H in G equal? If $gH = g'H$, then $g \in g'H$, so $g = g'h$ for some $h \in H$. In other words, $gH = g'H \iff (g')^{-1}g \in H$.

We denote the set of left cosets of $H \leq G$ by $G/H = \{gH : g \in G\}$. Does G/H carry a natural group structure?